Student dropout warnings

English presentation script

Main talk: slides 1-8 | 600 words in the spoken blocks | 4 suggested speakers

The spoken blocks take about 6.0 minutes at 100 words per minute, before pauses. Their main transitions are included. Rehearse aloud; the appendices are preparation material, not extra speech.

Speaker 1 | Slides 1-2

Slide 1: the question

Our question is whether information known at enrolment can identify later recorded dropout, and who the same warning rule misses. We are interested in student support. We first define the outcome, then evaluate overall prediction, and finally check how errors differ between groups. A warning does not determine a student's future. We therefore start by checking what the label means.

Slide 2: the outcome definition

We used 4,424 records from one Portuguese institution. We checked the original paper because dropout can mean different things. Here it includes changing course or institution, so it does not always mean leaving higher education. The outcomes were recorded at the normal end of the course, usually three years. We retained students still enrolled. In our binary analysis they have no recorded dropout at that point; they are not assumed to have graduated. These choices define what our predictions can mean.

Speaker 2 | Slides 3-4

Slide 3: what the entry profiles show

These plots use training records only. Recorded dropouts were older on average, but ages and admission grades still overlapped. Quiz 7 pointed us to Data to Viz. Its boxplot advice reminded us to show group sizes, so we added the counts below each outcome. We kept the boxes to compare typical values and spread. These observed patterns are associations, not causes.

Slide 4: inputs and evaluation design

We used age, admission grade and course, excluding later semester results and financial fields with unclear timing. We used course tools for summaries and plots. Logistic regression and validation were new to us, so we asked ChatGPT for explanations, checked R documentation, then rewrote and checked the main analysis. Training records fit the model. Validation records chose the threshold. Test records evaluated the frozen rule. Recorded gender was reserved for the main audit. A probability still needs a threshold before it becomes a warning.

Speaker 3 | Slides 5-6

Slide 5: the warning threshold

We wanted to find at least seventy per cent of dropout records in validation data while limiting false warnings. This illustrative target selected a common threshold of 0.31. It found 200 of 284 validation dropouts, about seventy per cent. We then froze the rule. The next slide checks test sensitivity and precision, and the cost in false warnings.

Slide 6: the actual test results

Sensitivity is the share of actual dropout records we find. Precision is the share of warnings that match a dropout record. Lowering the threshold raised test sensitivity from twenty-eight to sixty-six per cent, but reduced precision from sixty-four to forty-six per cent. We found 107 more dropout records and produced 177 more false warnings. Accuracy also fell. Ninety-eight dropout records were still missed. An overall average cannot tell us who those students were. We therefore checked the same errors by recorded gender.

Speaker 4 | Slides 7-8

Slide 7: whom the rule misses

Of the ninety-eight missed records, sixty-four were female and thirty-four male. We compare rates within actual dropouts: sixty-four out of 144 female records, and thirty-four out of 141 male records. That is forty-four versus twenty-four per cent, a gap of twenty percentage points. Its conditional bootstrap interval is roughly ten to thirty-one points. Male students had the higher false-warning rate. The five rows check coverage, both error types, and outcomes among both warned and unwarned students: independence, separation and sufficiency. Recorded dropout rates were also different, twenty-five versus forty-five per cent. These differences need investigation before real use; they do not prove discrimination.

Slide 8: the conclusion and next step

These errors explain why we would use a warning to start a conversation, with human review and a way to correct records. Before real use, we would check departure reasons, support capacity and performance on later cohorts at the intended institution. Our analysis found predictive signal in historical administrative records. We have not shown that acting on warnings prevents dropout. That would require a separate evaluation. Thank you.

Preparation notes

Use these explanations selectively when discussing the figures or answering questions. They are not additions to the timed talk.

Slide 2: why we presented the figure this way

We wanted the audience to see all three recorded outcomes, including students still enrolled. After checking the axis advice, we kept the existing bars: they already started at zero and gave both counts and percentages. We did not change a chart that already supported this comparison. Its colours distinguish categories; their order does not rank students.

Slide 3: why we presented the figure this way

We wanted a quick comparison of typical values and spread. Quiz 7 recommended Data to Viz, which led us to review these figures. The boxplot page pointed out that boxes can hide group sizes and distribution shapes. We kept boxplots for this limited comparison and added the training counts: Graduate 1,324, Enrolled 476 and Dropout 852. Equal-width boxes should not suggest equal numbers of students. We did not add thousands of raw dots to the main slide, and we do not claim the boxes reveal every distribution feature. The two panels use different, explicitly labelled units; we compare outcomes within each panel.

Slide 5: why we presented the figure this way

We wanted to show how three rates change as the numerical threshold moves. The axis discussion helped us distinguish this purpose from comparing bar lengths. We kept the lines and the full 0-100% scale. The colour caveats also reminded us to consider readers who cannot distinguish our colours. We gave the three curves different line patterns, so colour is no longer the only way to follow them. These are observed validation rates, not confidence bands; the selected threshold is still 0.31.

Slide 7: why we presented the figure this way

We wanted to compare error rates, but a group's total is not the denominator of every rate. The sample-size caveat prompted us to put the relevant n beside every point. For missed dropouts, those denominators are 144 female and 141 male dropout records, rather than all 575 and 314 students. We kept points with intervals because we are displaying binary rates and sampling uncertainty. The error-bar page stresses that bars can mean different things, so we explicitly identify ours as pointwise 95% percentile bootstrap intervals, conditional on the fixed model and threshold. They are not standard deviations or a guarantee about a future cohort. We also added circle and triangle markers. To compare the groups directly, we use the gap intervals in backup slide 12 rather than treating overlap of two separate intervals as a formal test.

Our reasons for key choices

Preparation material: use the reasoning relevant to a question; do not add every paragraph to the timed talk.

Slide 2: our reasoning

We kept Enrolled because our question concerns recorded dropout at the stated endpoint. Removing those students would change the population. We therefore call them no recorded dropout at that point, without claiming they graduated.

Slide 4: our reasoning

We want an enrolment warning, so a useful predictor must be known at enrolment. Later grades may predict well but answer a different question. We therefore keep a small set of defensible inputs and leave timing-uncertain fields out.

Slide 5: our reasoning

We use the seventy per cent validation target to make our trade-off explicit. It is our illustrative choice, not university policy. Once it selects a threshold, we keep that rule fixed so the test set can still provide a separate evaluation.

Slide 6: our reasoning

We cannot judge this warning by accuracy alone. Always predicting no dropout already gives 67.9% accuracy, while finding none of the recorded dropouts. We therefore report who is found and how many warnings are false, with separate denominators.

Slide 7: our reasoning

We compare missed-dropout rates within actual dropouts, because the groups have different sizes. The relevant comparison is 64/144 against 34/141. We also keep false warnings and both prediction strata, so our fairness discussion covers more than one favourable statistic.

Questions to practise

Which test counts support sensitivity and precision?

We found 187 of the 285 recorded dropouts, giving sensitivity of 65.6%. We issued 409 warnings; 187 matched a dropout record, giving precision of 45.7%. The other 222 warnings were false positives relative to the recorded endpoint. We missed 98 dropout records and correctly left 382 other records unflagged. We use these counts to explain the denominators rather than treating every percentage as accuracy.

Why use this dataset and this question?

The brief lists UCI as a source for alternative datasets. This release has an accessible source, numerical and categorical variables, and enough student records for a manageable project. Our question connects prediction to a concrete support concern. Dataset feasibility still needs the lecturer/course organiser's confirmation.

What exactly does a positive outcome mean?

Recorded Dropout at the source's normal course endpoint. The paper includes course and institution changes. It does not necessarily mean abandoning all higher education, and it is not a direct observation of who needs support.

Why retain students who are still enrolled?

The primary endpoint is recorded dropout by that point. Enrolled students have not been recorded as Dropout then, though they may leave later. The separate resolved-outcome check makes the consequences of excluding them visible; it cannot silently replace the primary population.

Why not use the very predictive second-semester grades?

They were not available at the chosen prediction time. They could support a different, later-stage question, but would not test an enrolment warning.

Why logistic regression, and is it in the supplied slides?

We had not used logistic regression in the supplied course material. We used AI explanations and official R documentation to learn how it models a binary outcome and returns probabilities, then reimplemented and checked the analysis. We treated course as categorical. We also learned that our age and grade terms are linear on the log-odds scale, with no interactions; these are assumptions we need to explain.

Why only three predictors?

They have a defensible enrolment interpretation and keep the model explainable. We chose scope over a broad feature search. This does not prove other entry variables are irrelevant or that this model is optimal.

Why three partitions?

We needed extra explanation of held-out evaluation. We learned to give each partition a different job: training fits coefficients, validation chooses the rule, and test estimates performance after both are fixed. We implemented this split in R and kept validation records out of the final fit. We did not use test results to choose the threshold.

Why seventy per cent?

We chose seventy per cent as an illustrative validation sensitivity goal. We learned how a threshold trades missed records against false warnings, then implemented a rule that maximised specificity among the eligible validation thresholds. We do not present our target as university policy or an objectively optimal choice. Real use would require consultation about benefits, costs and capacity.

What does AUC tell us?

We learned that AUC measures ranking rather than threshold-based accuracy. Our primary test AUC is 0.718, with a conditional bootstrap interval of 0.682 to 0.754. We interpret it as how often a randomly selected dropout record receives a higher score than a non-dropout record, counting ties halfway. We do not treat this as proof of calibration or usefulness at a particular threshold.

What does Brier score tell us?

We learned to evaluate the probabilities as well as the binary warnings. We used Brier score, the mean squared difference between predicted probability and the zero/one outcome; lower is better on comparable records. Our primary test value is 0.188, versus 0.218 for the training-prevalence baseline.

Why is female FNR higher even though gender was not an input?

Omitting a recorded attribute does not remove differences carried by other inputs, course composition or underlying outcome patterns. This descriptive audit identifies a difference. It does not isolate its cause or prove intentional discrimination.

Did adding gender solve the difference?

No. In this pre-specified comparison at the same threshold, AUC changed only slightly and the missed-dropout pattern became more unequal. That is why a small aggregate improvement cannot by itself determine a fair support rule. Both sets of results remain visible in the backup.

Can all three fairness criteria hold?

They condition on different denominators and may conflict when groups have different outcome rates and predictions are imperfect. We do not require all three to hold or make an unconditional impossibility claim. Which error matters depends on the use and the people affected.

What uncertainty does the bootstrap cover?

It resamples held-out student records within each recorded-gender group, with the fitted model and threshold held fixed. It describes variation in this conditional evaluation. It does not account for refitting the model, cohort clustering, transport to another institution or the effect of offering support. The intervals are pointwise, not simultaneous fairness certification.

Is this a future-cohort test?

No. The release has no cohort/year field, so our random hold-out mixes historical cohorts. Prospective or chronological validation is an important missing step.

What is still needed from the group?

Review the data definitions and code, obtain the required topic-feasibility confirmation, agree the speakers, practise aloud, attend check-ins, set up actual repository collaboration and complete the separately required report, peer evaluation and submissions. None is claimed completed by the slide files.

What assistance was used?

We asked ChatGPT to explain unfamiliar methods and used its help with initial code and text drafts, source checks and slide formatting. We then rewrote and ran the main R analysis and checked the outputs. R generated the statistical results; we are responsible for our final interpretation.

Figure-design references

Data to Viz: The Boxplot and its pitfalls - https://www.data-to-viz.com/caveat/boxplot.html

Data to Viz: The issue with error bars - https://www.data-to-viz.com/caveat/error_bar.html